

Download Video

Easily download videos from Youtube and other sites.

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Abstract

Download Video is a commandline application for downloading videos from Youtube and other sites. The user can select a video format to download based on one or more parameters, such as video height, whether to include audio, et. al.

Part I

Terminology

Video In this document, refers to any video or audio file, including videos with or without audio.

Site processor A singleton class that contains methods for handling site-specific implementations of the **SiteProcessor** interface.

Part II

User Interface

Chapter 1

DONE Main Application

- State "DONE" from "TODO" [2025-06-21 Sat 21:34]

Download Video uses a CLI to interact with users. The basic syntax is this:

Usage:

```
download-video [-hny] URL
download-video [-hny] [--height HEIGHT] [--video-only] URL
download-video [-hny] [--audio-only] URL
download-video [-hny] -r FILE
```

The lone positional argument is treated as a string URL by default; if **-r** is passed, it's treated as a path.

Options

Short	Long	Note
-n	--dry-run	Prints only what will happen
-r	--rename	Renames the file instead of downloading from <i>URL</i>
	--height <i>NUMBER</i>	Selects the best format according to the video height.
	--audio-only	Selects only audio formats
	--video-only	Selects only video formats
-y	--yes	Automatically says yes to every prompt

Chapter 2

Config

This, shall we say, subcommand is derived from a submodule of the main package. It has the following CLI:

Usage:

```
download-video-config [-h] VARIABLE [VALUE]
```

If *VALUE* is provided, then *VARIABLE* is set to that value. Otherwise, the current value of *VARIABLE* is shown.

Part III

Technical Specification

Chapter 3

Dependencies

Download Video uses:

typer CLI

yt-dlp Backend

pydantic Validation of data, primarily the info dictionary

Standard library:

urllib.parse It contains urlparse, which we use to normalize URLs

Chapter 4

TODO Overview

- State "DONE" from "TODO" *[2025-06-21 Sat 21:31]*

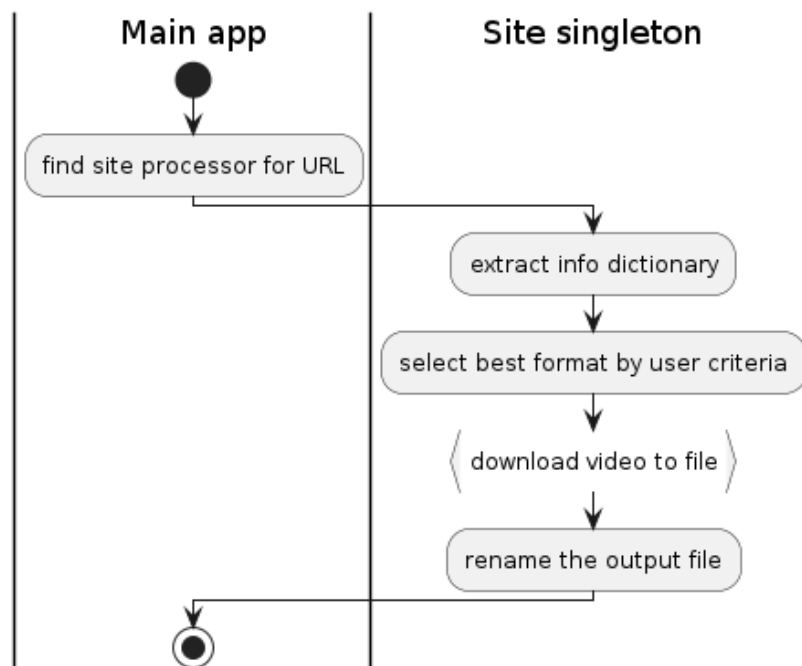


Figure 4.1: Overview

4.1 DONE Getting Site Processor

- State "DONE" from "TODO" [2025-06-21 Sat 22:42]

The URL is matched with a list of preregistered singletons known as *site processors*. That is to say, certain Python modules are treated as singletons. When one of them matches with the URL, it is used for the remainder of the process.

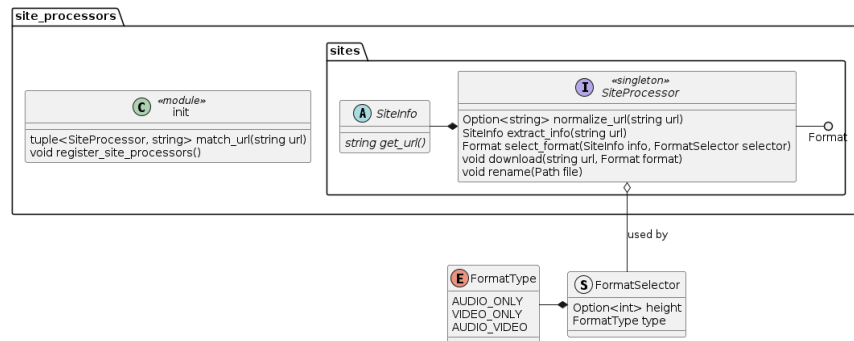


Figure 4.2: Site Processor

4.2 DONE URL Normalization

- State "DONE" from "TODO" [2025-06-21 Sat 22:48]

The site processor has a function for normalizing URLs, which is done before extracting the info dictionary. The exact method will change depending on the site processor (see Supported Sites).

4.3 TODO Extracting Info Dictionary

To extract an info dictionary, normally you would have to do this:

```
params = {
    'clean_infojson': True,
    'restrictfilenames': True,
}
with YoutubeDL() as ydl:
    info = ydl.sanitize_info(ydl.extract_info(url, download=False))
```

Now, this is done in the selected site processor. Exactly what structure the info dictionary has will depend on the site the URL belongs to. A `SiteInfo` object is returned to the caller, which then passes that `SiteInfo` to the next step of the process.

4.4 TODO Selecting A Format

The `select_format` method selects a suitable format based on the criteria provided by the user. This criteria comes in the form of a `FormatSelector` class.

The exact algorithm used for selecting a format depends on the site processor used. And the format itself is an opaque type (likely a string) whose exact value we cannot know from the outside. This is represented in Figure 4.2.

But in general, the following process is followed:

1. From the info dictionary, a list of formats is extracted.
2. The format list is filtered to only include audio and video formats, including audio/video formats.
3. The format list is split three lists:
 - A. Audio-only formats
 - B. Video-only formats
 - C. Audio/video formats
4. The format list is filtered using the strategy pattern. See each supported site for details.

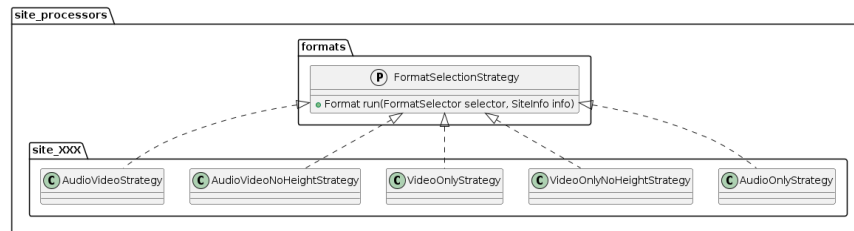


Figure 4.3: Format Selections

Chapter 5

Configuration

A configuration file controls the exact behavior of Download Video. There is a global section followed sections for each supported site. Their respective values are explained under Supported Sites.

The `config` subpackage handles the entire configuration framework. In the standard config directory, a Toml file is used called `config.toml`.

Chapter 6

Supported Sites

6.1 TODO Youtube

Right now, Youtube is the only site supported by Download Video. This will likely change in the future when this project has taken shape and is out of alpha.

The site processor for Youtube matches URLs with the following formats:

- `https://www.youtube.com/watch?v=XXXXXXXXXXXX`
- `youtu.be/XXXXXXXXXXXX`
- `yt:XXXXXXXXXXXX`

This site processor normalizes the URL so that it always has the first form. The info dictionary returned by Youtube has this structure:

1. **TODO** URL Normalization URL normalization is done by matching a URL with a regular expression and capturing the video ID, then returning a "proper" URL with said video ID.
2. **TODO** Format Selection Selecting a format is done by trying to get the best (highest quality) video which satisfies the criteria outlined in `FormatSelector` (see Figure 4.2).

A series of strategies are used to filter the list formats according to the criteria given. The name of each strategy class pretty much says what the strategy will be. Then the highest quality format is chosen. The barometer for a format's quality is its total bitrate, denoted in the `tbr` element.

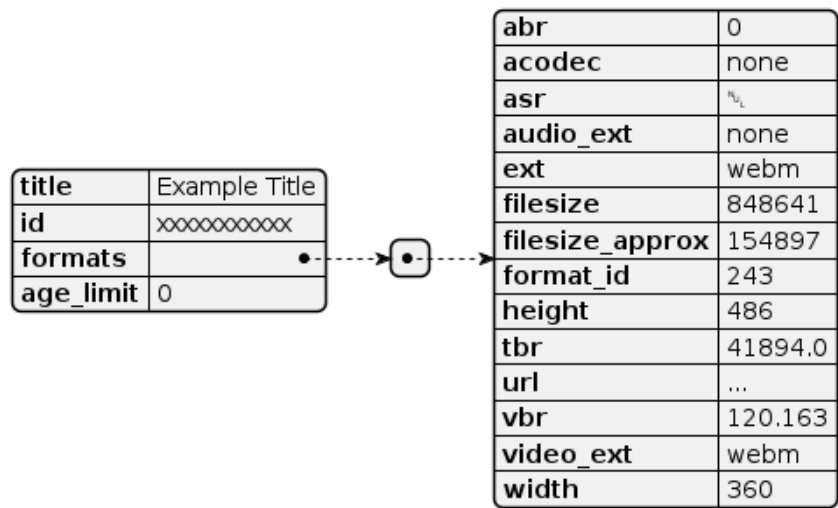


Figure 6.1: Info Dictionary - Youtube

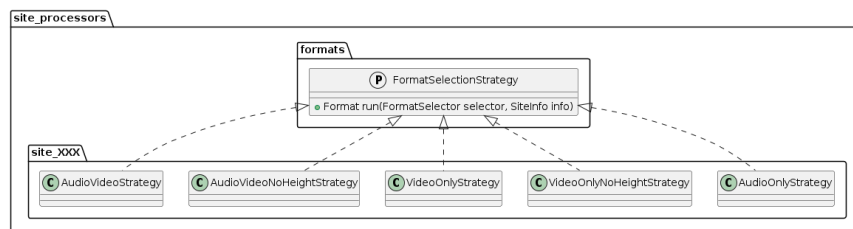


Figure 6.2: Format Selections